

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #28: Inter-City Freight Load Matching Marketplace

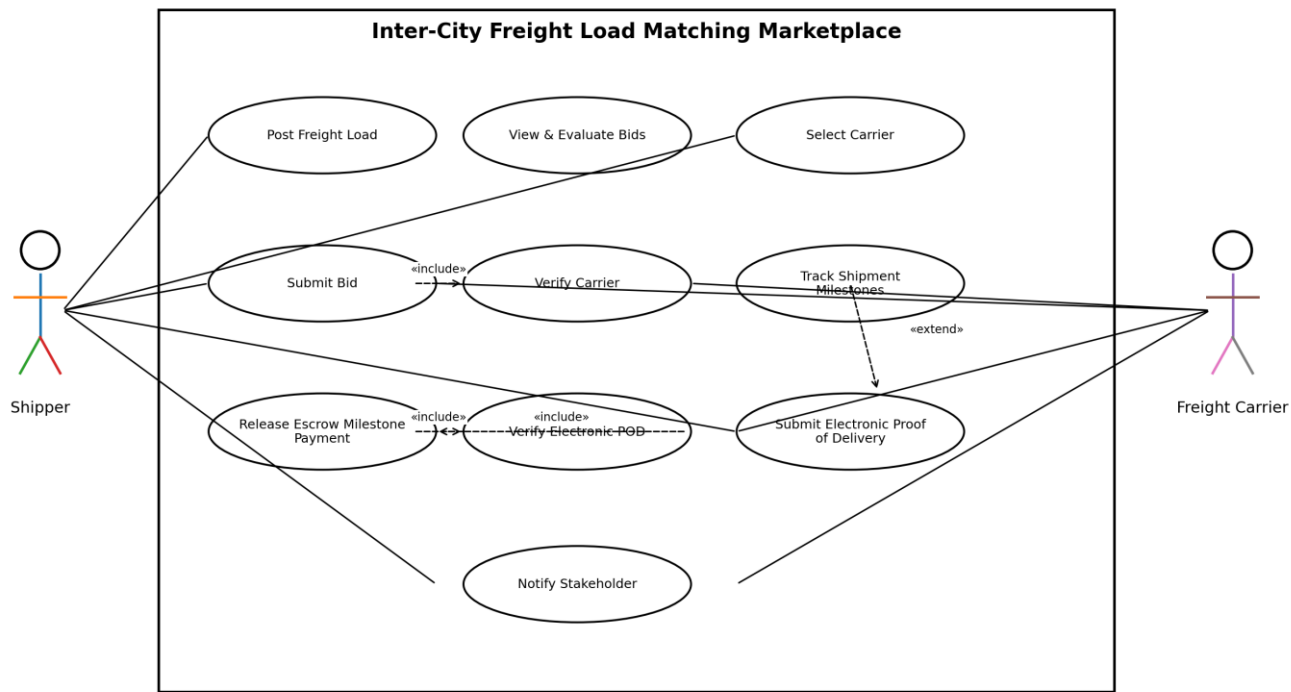
Smart Cities, Transport & Logistics

1. Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a shipper to post a freight load with the shipment details and a specified auction timeframe.	High	Pass: Load is recorded and becomes visible to eligible carriers during the auction timeframe. Fail: Load is posted without required shipment or auction information.	Load posting is the starting point of the freight marketplace and provides the shipment that carriers can bid on.
FR-002	Functional	The system shall allow verified trucking carriers to submit competitive bids on posted freight shipments within the specified auction timeframe.	High	Pass: Valid carrier bid is recorded and the shipper is notified. Fail: An unverified carrier or a bid submitted after the auction closes is accepted.	Directly supports the core load-matching activity and follows the supplied FR guideline.
FR-003	Functional	The system shall allow the shipper to review submitted bids and select a carrier for a freight shipment.	High	Pass: Shipper can view eligible bids and a selected carrier is recorded. Fail: An invalid or unsubmitted bid can be selected.	A shipper needs a controlled mechanism to compare carrier offers and award the shipment.
FR-004	Functional	The system shall allow shipment milestones and electronic proof of delivery (e-POD) to be recorded and verified.	High	Pass: Milestone/e-POD data is recorded and the e-POD can be verified before payment release. Fail: Payment workflow proceeds from an unverified e-POD.	The marketplace depends on milestone completion and proof-of-delivery sign-off to trigger payment.
FR-005	Functional	The system shall automatically release the applicable milestone payment from escrow after verified electronic proof of delivery is submitted.	High	Pass: Eligible escrow payment is released after verified e-POD and the transaction status is updated. Fail: Payment is released without verified e-POD.	Automated milestone payment release is a defining business function in the problem statement.
NFR-001	Non-Functional: Performance & Security	Milestone payments held in escrow shall disburse automatically within 60 seconds of verified electronic proof-of-delivery submission.	High	Pass: Benchmarking tests confirm payment disbursement within 60 seconds under simulated peak load while required security controls remain satisfied. Fail: The target latency or security requirements are not met.	This directly follows the supplied NFR guideline and establishes measurable payment responsiveness and security.
NFR-002	Non-Functional: Reliability	The system shall maintain transaction integrity so that a freight bid, carrier selection, milestone update, e-POD verification, and escrow payment release are recorded without loss or duplication.	High	Pass: Reliability tests show that completed transaction events remain consistent after normal and peak-load operations. Fail: an event is lost, duplicated, or causes an inconsistent payment state.	Freight matching and escrow payment require trustworthy records; inconsistent transaction state could affect both parties.

2. UML Use-Case Diagram

The diagram models the two target actors identified in the problem statement: Shipper and Freight Carrier. It includes mandatory «include» relationships and a conditional «extend» relationship.



Note: «include» shows mandatory reused behavior; «extend» shows optional/conditional behavior.

3. Use-Case Flow Specification

Use Case: UC-02: Submit Competitive Freight Bid

Primary Actor: Freight Carrier

Supporting Actor: Shipper

Goal: Allow a verified carrier to submit a competitive bid for an available freight load.

Preconditions: 1. A freight load has been posted by a shipper.

2. The load is within its specified auction timeframe.

3. The carrier has a valid verified status and required permits.

Postconditions: 1. The bid is stored against the selected freight shipment.

2. The shipper is notified that a new bid has been received.

3. The bid remains available for the shipper to review until the auction process permits selection.

Main Success Scenario: 1. Carrier views available posted freight loads.

2. Carrier selects an eligible load.

3. System displays shipment details and the auction closing information.

4. Carrier enters the proposed bid details.

5. System verifies that the carrier is verified and eligible to bid.

6. System checks that the auction timeframe is still open.

7. System records the bid with its submission time and carrier details.

8. System confirms successful bid submission to the carrier.

9. System notifies the shipper of the new bid.

Alternate Flow: A1. At Step 5, the carrier is not verified or does not have valid permits. The system rejects the bid, explains that verification is required, and does not expose the invalid bid to the shipper.

Relationships: «include»: Submit Bid includes Verify Carrier because eligibility verification is mandatory before accepting a bid.

«extend»: Track Shipment Milestones may be extended by Submit Electronic Proof of Delivery when the shipment reaches the delivery stage.

Submission Notes

This package is based on the uploaded Problem Statement #28. The supplied statement explicitly identifies Shipper and Freight Carrier as target stakeholders/actors, gives the bidding requirement as FR guidance, and specifies the 60-second escrow disbursement behavior as NFR guidance. The remaining requirements and use-case details are derived to complete the requested Lab 1 deliverables while keeping the marketplace's core workflow consistent with the statement.